Yang XIAO

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Education Background

Institut de Mathématiques de Marseille, Aix-Marseille Université, Marseille, France 10/2024-present

- Doctorant in Mathematics Supervisor: Prof. Colin Guillarmou & Rémi Rhodes
- PhD Project: Towards conformal bootstrap of Compactified Imaginary Liouville Theory
- Expected defense: by June 2027

School of Mathematical Sciences, Fudan University (FDU), Shanghai, China 09/2021-06/2024

- M.S. in Probability, GPA: 3.9/4.0 Supervisor: Prof. Elie Aïdékon
- Main Courses: Modern Probability, Stochastic Analysis, Large Deviation, Functional Analysis, Modern PDE, Complex Analysis, Differential Geometry, Malliavin Calculus

Honor National Scholarship (top 10%)

School of Mathematics, Shandong University (SDU), Shandong, China 09/2018-06/2021

- B.S. in Statistics, GPA: 4.74/5.0

School of Computer Software, SDU, Shandong, China 09/2017-07/2018

Honor Academic Scholarship for Outstanding Students (top 5%)

Research Projects

- Y. Xiao, Y. Xie. **Segal's semigroup, Virasoro structure, and Ward identities in Compactified Imaginary Liouville Theory**. Preprint, arXiv:2610.22170.
- Y. Xiao, Y. Xie. **Boundary Compactified Imaginary Liouville Theory**. Preprint, arXiv:2511.11269.
- D. Qian, Y. Xiao. **Speed of random walk in Dirichlet environment on a Galton–Watson tree**. To appear in Ann. Inst. Henri Poincaré Probab. Stat. arXiv:2405.11845.
- D. Qian, Y. Xiao. **Scaling limit of the random walk on a Galton–Watson tree with regular varying offspring distribution**. To appear in Electron. Commun. Probab. arXiv:2309.09200.

Ongoing project:

- With C. Guillarmou, A. Kupiainen, R. Rhodes, and Y. Xie: **Bootstrap of Compactified Imaginary Liouville Theory,**
- **Virasoro orbital measure from semiclassical limit of annulus Liouville CFT,** in preparation.

Talks

- "Conformal structure of Compactified Imaginary Liouville Theory", talk at Loop Models, Conformal Field Theory, and Geometry, CIRM, Marseille, France, August 2026.
- "Towards solvability of Compactified Imaginary Liouville Theory", seminar talk at Fudan University, Shanghai, China, January 2026.
- "Speed of Random Walks in Dirichlet Environment on a Galton-Watson Tree", talk at YMSC Probability Seminar, Tsinghua University, Beijing, China, April 2024.